

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Computer Vision with OpenCV COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO4: Evaluate recognition and learning methods including machine learning and deep learning models (CNNs, GANs, SVM, HMM, etc.) for vision tasks.. (BL5)

Assessment Tool Weightage: 20%

QUESTIONS: 2 Total Marks:20 Annexure A

Comparative Analysis

Code of Conduct:

I VALLAPU MOHAN KUMAR (192524225) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V. Mohan Kumar

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Selection of Studies/Parameters	20%	Selects highly relevant and diverse studies with well-defined comparison parameters	Selects relevant studies with minor gaps in parameter definition	Limited or partially relevant selection	Poor or inappropriate selection of studies	
Comparison & Differentiation	25%	Clearly compares multiple aspects (methods, results, limitations) with strong differentiation	Good comparison with minor gaps	Basic comparison with limited depth	No clear comparison; mostly descriptive	
Critical Evaluation	25%	Critically evaluates strengths, weaknesses, and implications with strong justification	Provides evaluation with some justification	Limited evaluation; lacks depth	No critical evaluation provided	

Synthesis & Insight Generation	20%	Generates meaningful insights and conclusions from comparisons	Some insights derived from comparison	Limited or unclear insights	No insights generated	
Clarity	10%	Information is clearly presented (tables/figures) with logical structure	Generally clear with minor issues	Presentation lacks clarity or structure	Poor presentation and difficult to understand	

Annexure A

Comparative Analysis

1. **CNNs vs. Vision Transformers (ViTs) for Object Detection Scenario:** A self-driving car company needs to detect rare urban obstacles (traffic cones, animals) in real-time using limited labeled datasets.

Questions:

1. Compare CNNs and Vision Transformers in terms of:
- Feature extraction capabilities for local vs. global context

○ Performance with small or sparse objects

○ Computational complexity and inference speed for embedded systems

Answer:

Parameter	CNNs	Vision Transformers (ViTs)
Feature Extraction	Excellent at extracting local features such as edges, corners, and textures using convolution filters.	Captures global context using self-attention, understanding relationships between distant image regions.
Performance with Small/Sparse Objects	Better for detecting small objects because of local receptive fields and multi-scale feature maps.	May miss small objects unless trained with large datasets or enhanced architectures.

Parameter	CNNs	Vision Transformers (ViTs)
Computational Complexity	Lower computational cost and memory usage.	Higher computational complexity due to self-attention operations.
Inference Speed	Faster inference, suitable for embedded devices.	Slower inference and requires more computational resources.
Training Data Requirement	Performs well with moderate-sized datasets.	Requires very large datasets for best performance.

2. Discuss which method is better suited for real-time deployment and how transfer learning or data augmentation could impact performance.

Answer:

CNNs are better suited for real-time deployment in self-driving cars because:

- They have faster inference speed.
- They require less memory and computational power.
- They perform well on embedded hardware such as NVIDIA Jetson devices.
- They accurately detect small objects like traffic cones and animals.

Role of Transfer Learning

- Uses pre-trained models such as ResNet, YOLO, or EfficientNet.
- Reduces training time.
- Improves accuracy when labeled data is limited.

Role of Data Augmentation

- Increases training samples using rotation, flipping, scaling, cropping, brightness adjustment, and GAN-generated images.
- Improves robustness.
- Reduces overfitting.
- Enhances rare object detection.

Conclusion:

For limited datasets and real-time object detection, **CNNs with transfer learning and data augmentation** provide the best balance between speed and accuracy.

2. PCA-Based Shape Priors vs. Graph-Based Learning for 3D Shape Correspondence

Scenario: An animation studio is aligning 3D human face meshes for morphing applications. Meshes vary in topology and vertex density.

Questions:

1. Compare PCA-based shape priors and graph-based learning in terms of:
 - Ability to handle non-rigid deformations
 - Robustness to varying mesh resolutions and noise
 - Ease of implementation and computational cost

Answer:

Parameter	PCA-Based Shape Priors	Graph-Based Learning
Handling Non-Rigid Deformations	Limited capability; mainly models linear shape variations.	Excellent; captures complex non-rigid deformations effectively.

Parameter	PCA-Based Shape Priors	Graph-Based Learning
Robustness to Mesh Resolution & Noise	Sensitive to varying mesh resolutions and noise.	Highly robust to different mesh structures, resolutions, and noisy data.
Ease of Implementation	Simple mathematical implementation using dimensionality reduction.	More complex; requires graph neural networks and graph processing.
Computational Cost	Low computational cost and faster execution.	Higher computational cost due to graph computations.
Accuracy	Moderate accuracy for simple shapes.	High accuracy for complex 3D face correspondence.

2. Analyze which approach would provide more accurate correspondences for realistic morphing and why.

Answer:

Graph-Based Learning provides more accurate correspondences because:

- It preserves neighborhood relationships between mesh vertices.
- Handles complex facial expressions and non-rigid deformations.
- Performs well even when mesh topology and vertex density differ.
- Produces realistic face alignment for morphing applications.
- Is more robust against noise and irregular meshes.

Advantages

- Higher correspondence accuracy.
- Better feature representation.

- Suitable for realistic animation and face morphing.
- Generalizes well to unseen mesh structures.

Disadvantages

- Higher computational cost.
- Requires larger datasets and longer training time.

Conclusion:

For realistic 3D face morphing, **Graph-Based Learning** is the preferred approach because it generates more accurate and reliable shape correspondences than PCA-based shape priors, despite its higher computational requirements.